

Healthy teams achieve better results.

AI –Based Employee Wellness Management Platform

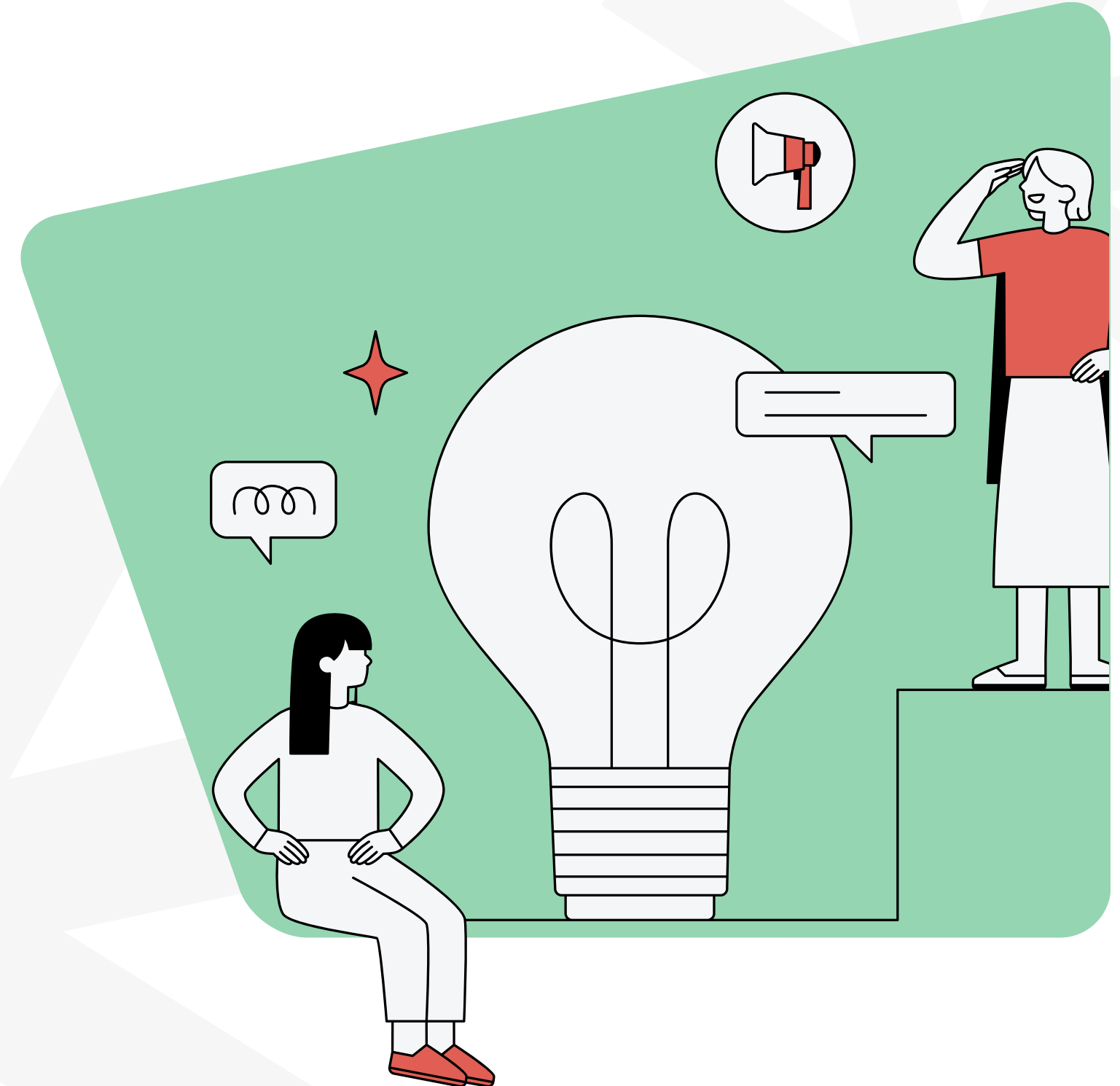
Support Employee's Mental Health and Sentiment Remotely

Presented By:

Team 4



**Rishit Ranjan
Ritheesh Reddy
Sanjana Teggi
Simran Hakim
Shivendra Tripathi
Sudip Madhu**



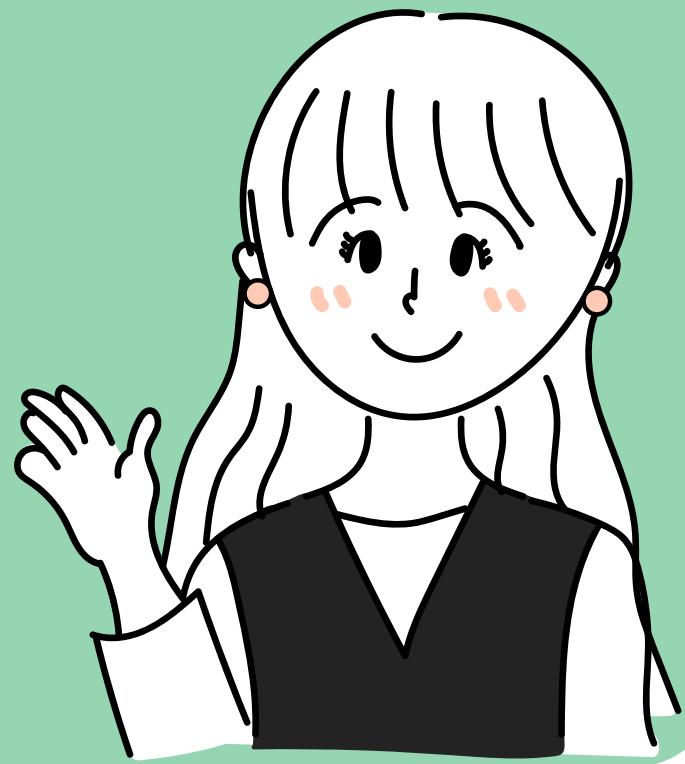
Problem Statement



→ Current System's Limitations and Issues

- Lack of a secure and centralized login and wellness management system.
- Difficulty in managing user/admin access and employee wellness records.
- Limited real-time analysis of workforce performance and health trends.
- Need for instant wellness guidance through an intelligent chatbot with natural language response.





Introduction

→ **Knowing about the Project**

- Employee wellness is essential for strong productivity and steady growth. The system helps manage health data and spot risks like stress while offering personal wellness tips.
- Employee wellness plays a big role in performance and growth. This system handles health data, predicts risks, and gives each person wellness ideas that support their wellbeing.
- Supporting employee wellness leads to better focus and stronger results. The system works with wellness data, identifies stress risks, and shares recommendations and the sentiment to help everyone stay healthy.

Project Objective



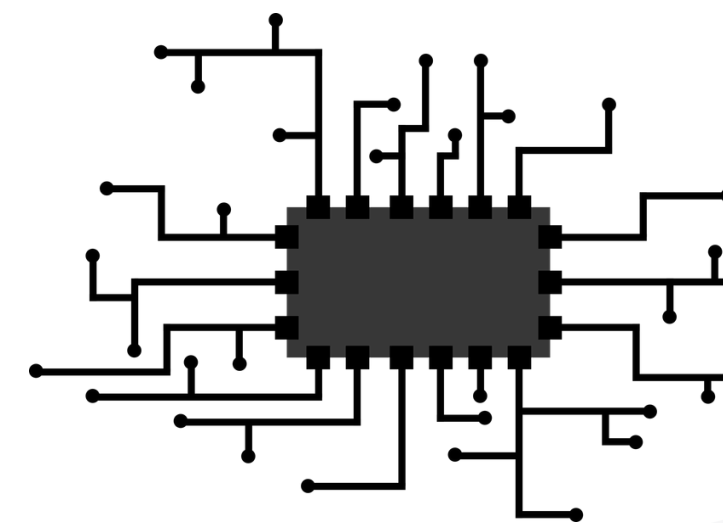
→ Objectives of the Project

1. Personalized Employee Dashboards: Tracks vitals, sets goals, and delivers custom AI insights.
2. Proactive Admin Analytics: Enables predictive risk modeling, sentiment tracking, and key KPIs.
3. 24/7 AI Wellness Assistant: Provides instant, contextual advice on diet, stress, and exercise.
4. Secure & Scalable Infrastructure: Ensures data privacy, role-based access, and centralized analytics.
5. Automated Admin & Emergency Response: Streamlines claims, scheduling, and instant SOS alerts.



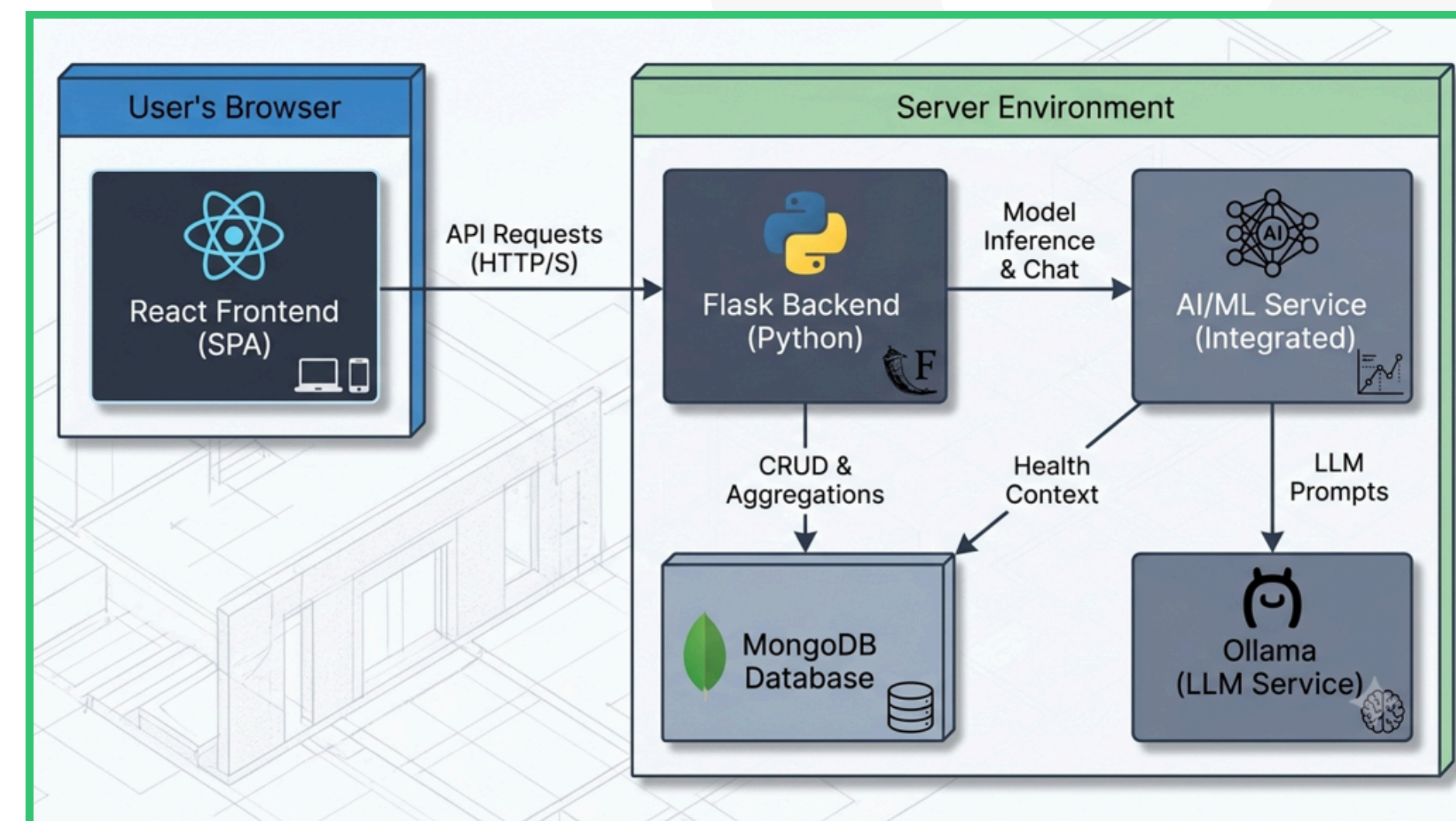


System Architecture

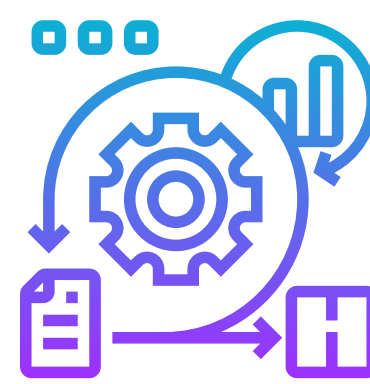


Explaining the structure and flow of the system.

- The application is built on a modern, decoupled architecture, commonly known as a Backend for Frontend (BFF) pattern.
- This means the React frontend is a distinct application that communicates with a dedicated Flask backend via a REST API
- Microservice-Ready AI Integration: Isolates LLM interactions and machine learning model inference into modular background services to maintain high responsiveness and system efficiency.
- Stateless Token Authentication: Utilizes secure JWT authorization across API requests, ensuring strict role-based access control and seamless horizontal scaling.

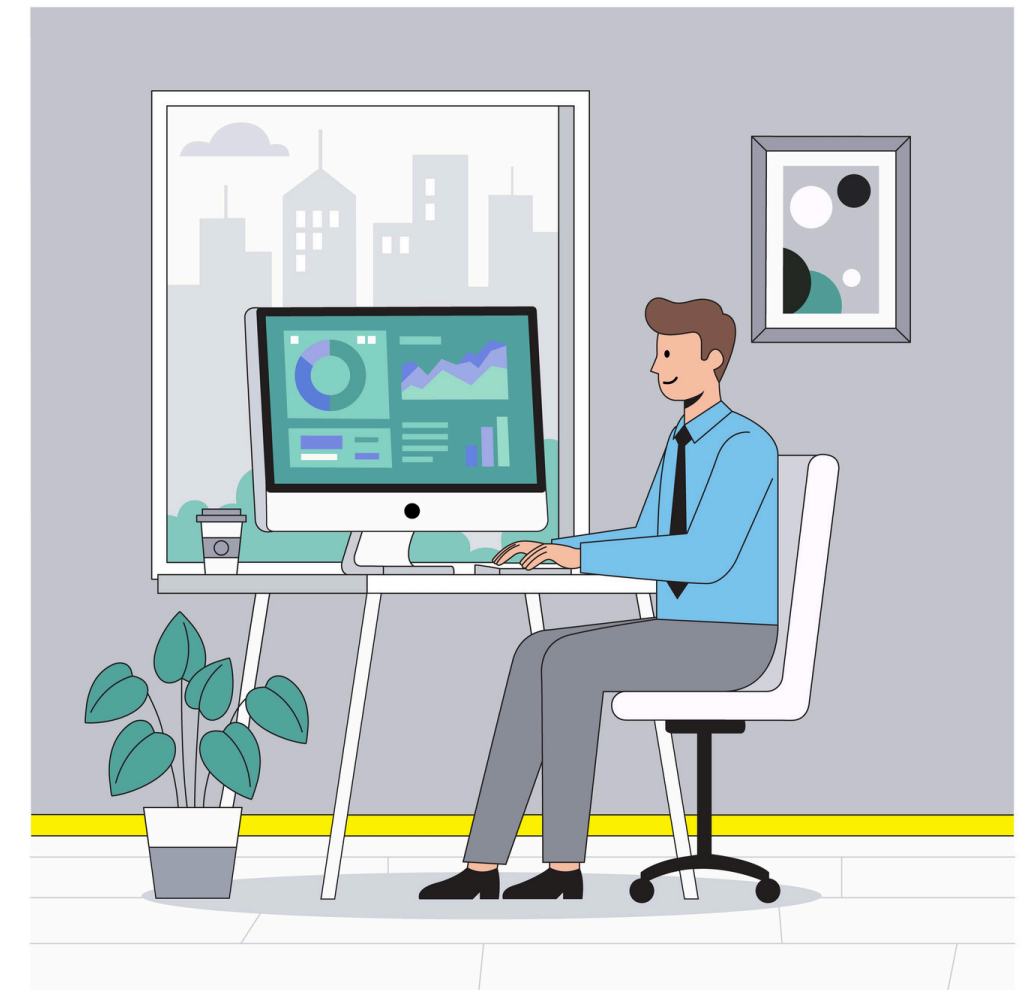


Methodology



→ How the system was developed and works step-by-step.

- The system was developed in a modular way, starting with secure user authentication and role-based access for employees and admins.
- After that, the frontend, backend, and database were connected to manage wellness records, analytics, and user interactions smoothly.
- Finally, AI-based features such as risk prediction, recommendations, sentiment analysis, performance analytics, and chatbot support were added to make the platform smarter and more useful.



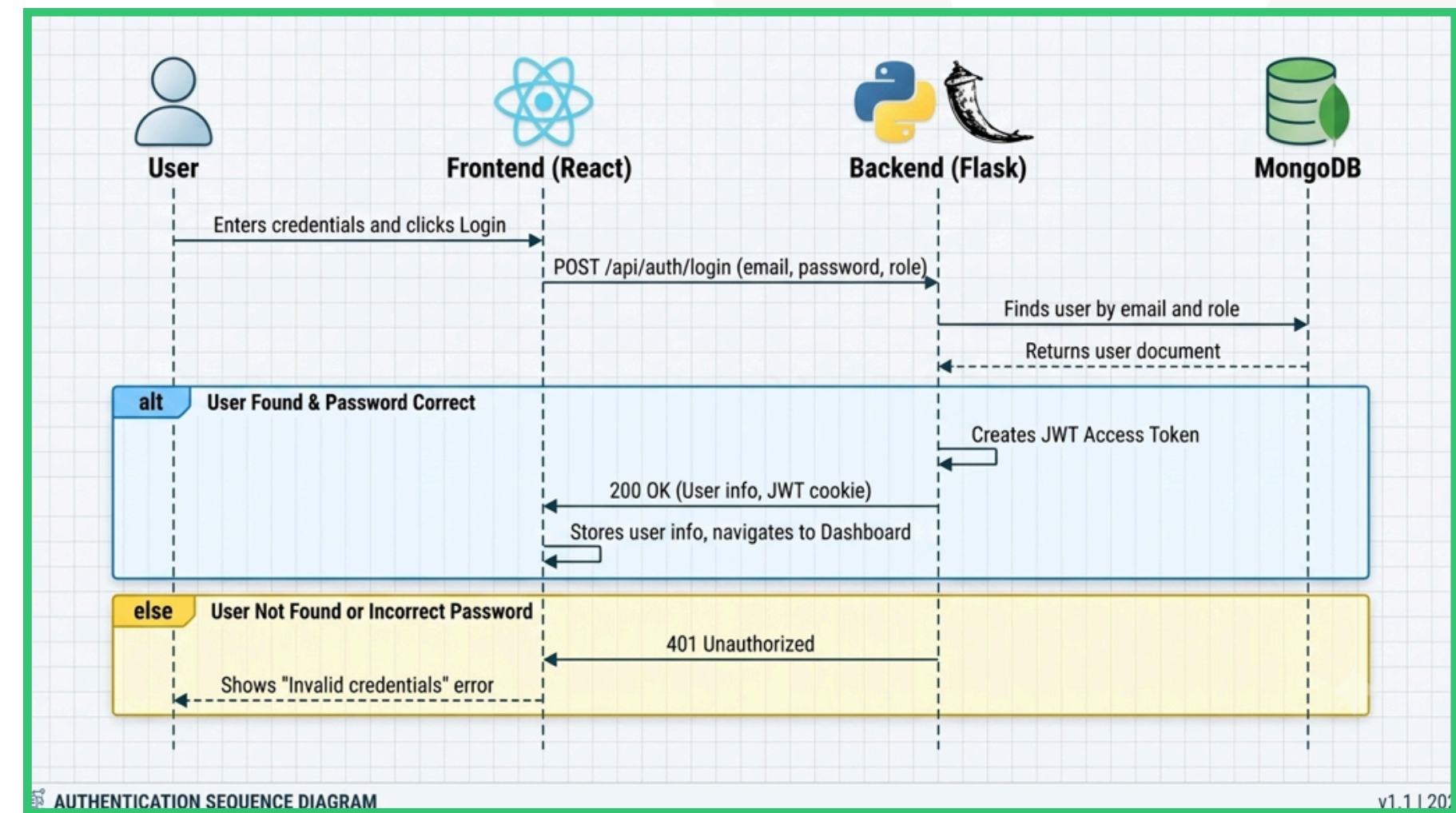


User Authentication Flow



→ Employee Journey

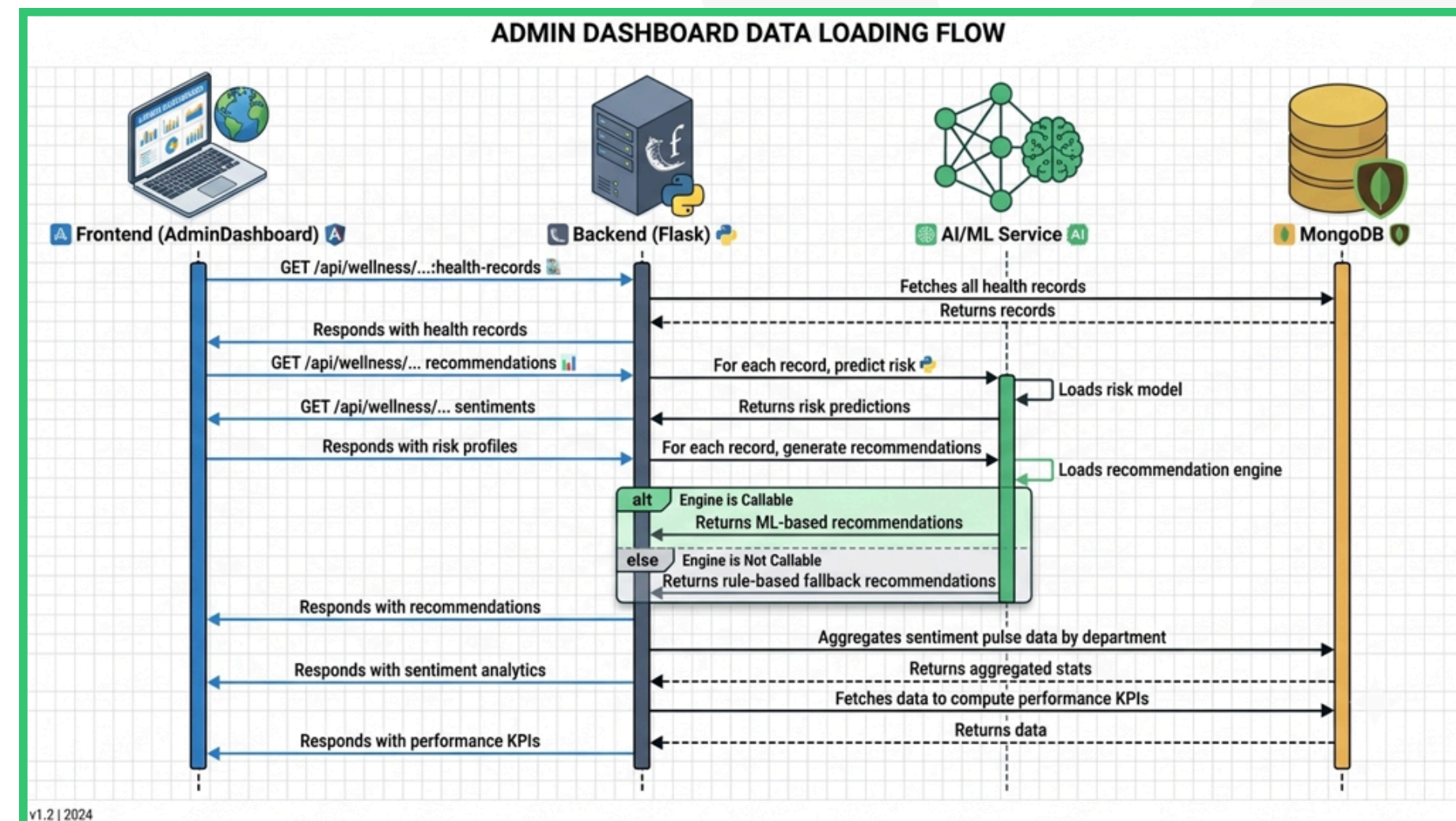
- Auth & Profile: User registers/logs in using JWT authentication → Dashboard loads.
- Data Input: User logs daily habits, sleep, exercise, and mental health feedback.
- Interaction: User requests personalized meal/fitness plans or chats with the InfyWell AI Assistant.
- Action & Emergency: User books checkup appointments or triggers an Emergency SOS alert.



Admin Flow

→ HR/Management Overview

- Dashboard Access: Admin authenticates → Views organization-wide health records and department filters.
- Analytics Monitoring: Admin monitors burnout risk trends, absenteeism rates (3.6%), and overall workforce risk metrics.
- System Management: Admin adjusts risk thresholds (e.g., High Risk = 70), configures privacy settings, manages insurance claims, and sends broadcast notifications.
- Emergency Handling: Admin receives real-time SOS alerts and health checkup logs.

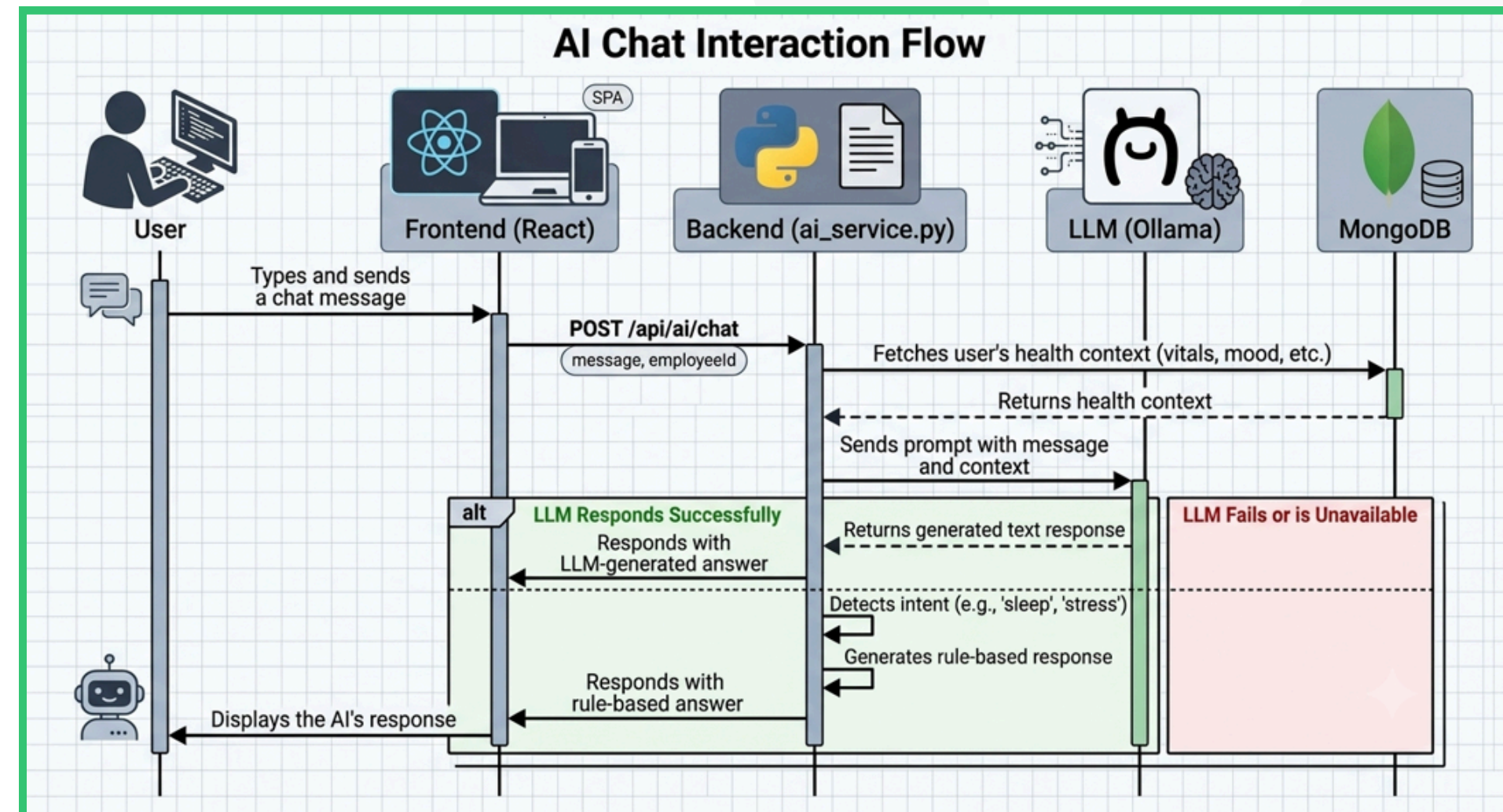


AI Chat Interaction Flow

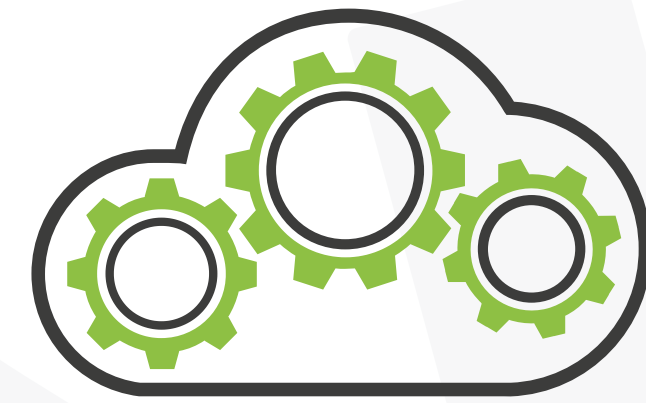


Data Processing Engine

- Data Ingestion: Collects raw health metrics, BMI, blood pressure, and text feedback from MongoDB.
- Inference Pipeline:
- ML Models (Random Forest/Decision Tree): Calculates risk severity scores (<45% Low, 45-60% Moderate, >75% High).
- NLP Engine: Performs sentiment and emotion analysis on employee feedback/logs.
- Ollama LLM Service: Ingests health context and user prompts to generate personalized daily tips, diets, and conversational advice.
- Output Delivery: Renders predicted risks and recommendations back to React Frontend dashboards in real time.

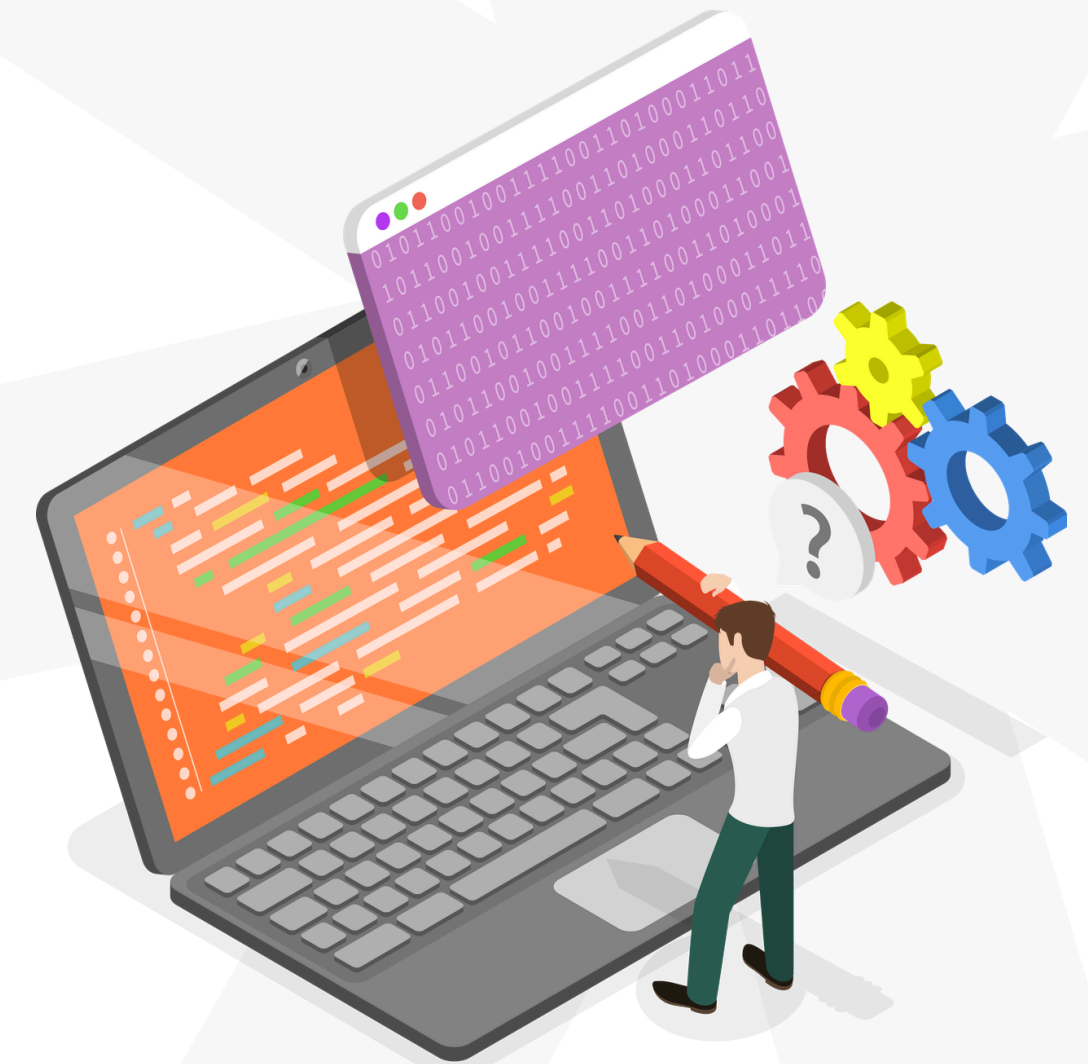


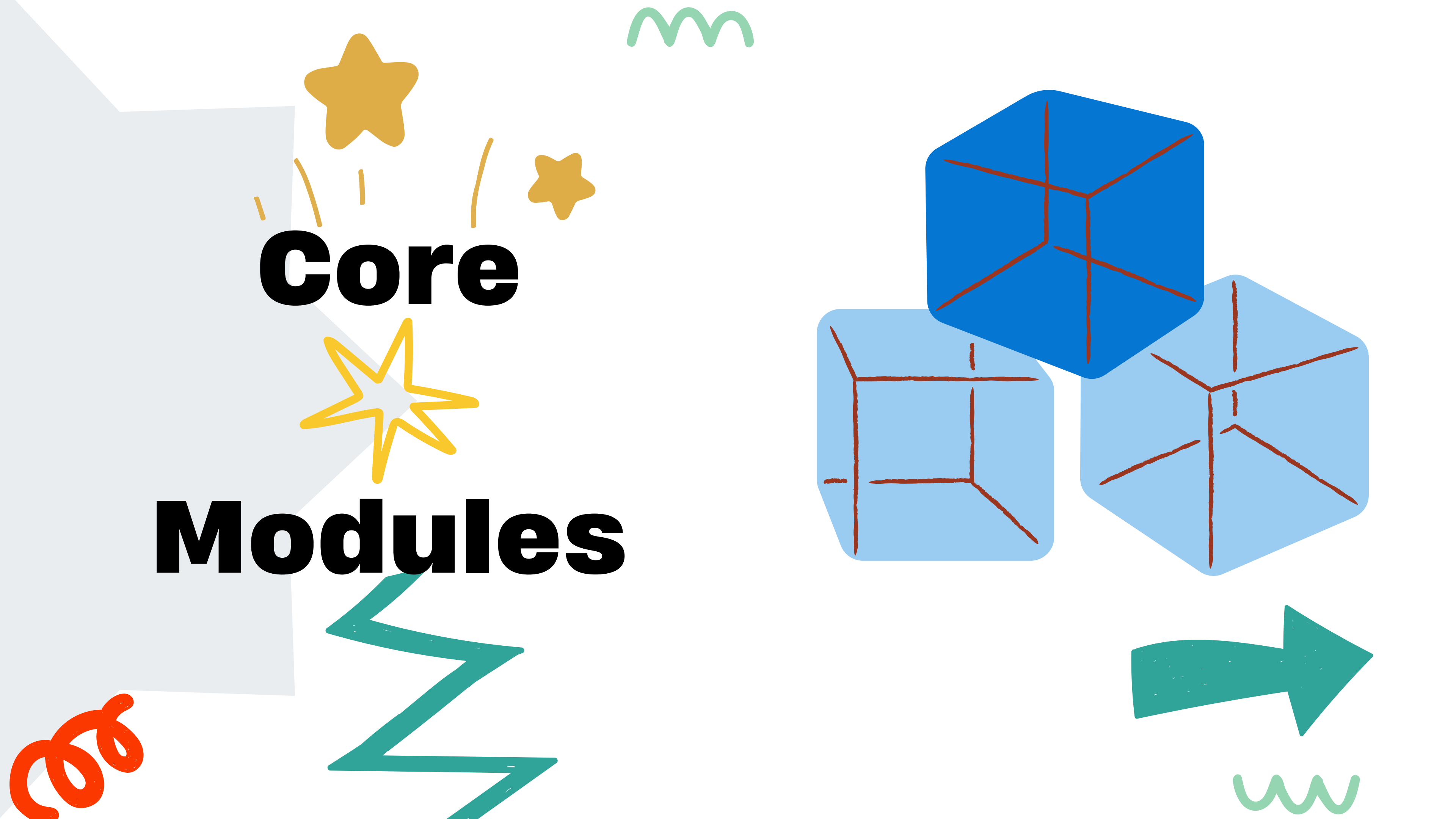
Software Specification



→ Software Requirements to run on local machine

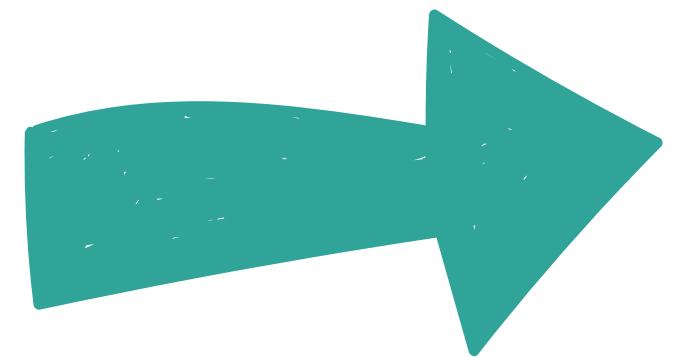
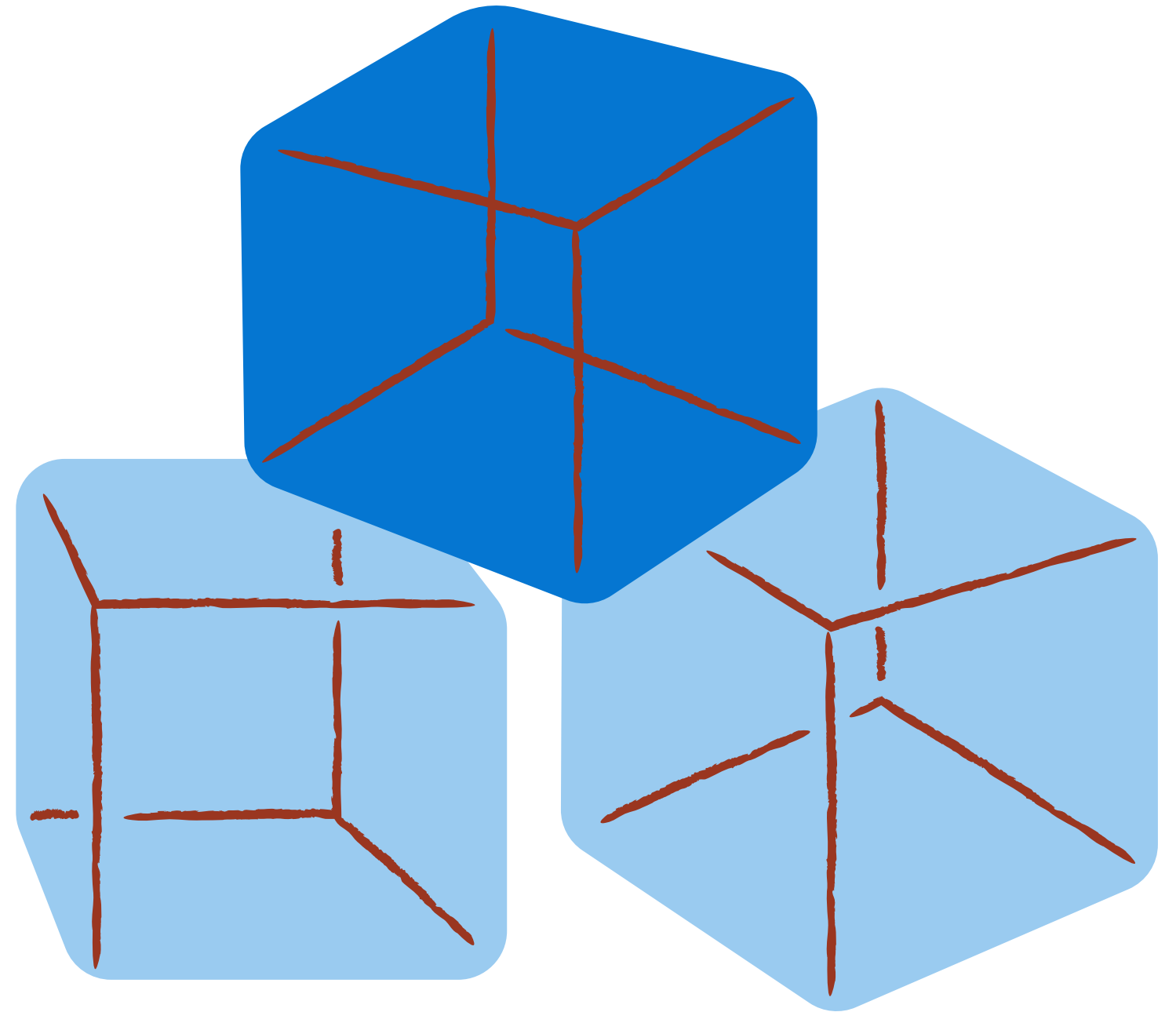
- Flask API with Python and NodeJs.
- Database: MongoDB.
- Authentication: JWT and bcrypt for secure login and password handling.
- Tools/Libraries: React Router, Flask-CORS, PyMongo, and ML libraries for wellness analytics.





Core

Modules





Login & SignUp Module



SignUp and SignIn

- User login with email and password.
- User registration with required fields.
- Remember me option in login form.
- Forgot password flow with email, OTP, and reset password steps in the frontend prototype.
- Role-based redirect to user or admin dashboard after authentication.



Create your account

Deploy wellness metrics and trackers for your team

FULL NAME

 John Doe

WORK EMAIL

 john.doe@company.com

PASSWORD

 At least 6 characters 

CONFIRM PASSWORD

 Match password 

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#).

Create Account →

Already have an account? [Sign In](#)



Employee Sign In

Use your registered employee credentials.

SELECT YOUR ROLE

 Employee

 Admin

EMPLOYEE ID

EMP101

EMAIL ADDRESS

 rishitranjan131@gmail.com

PASSWORD

[Forgot password?](#)

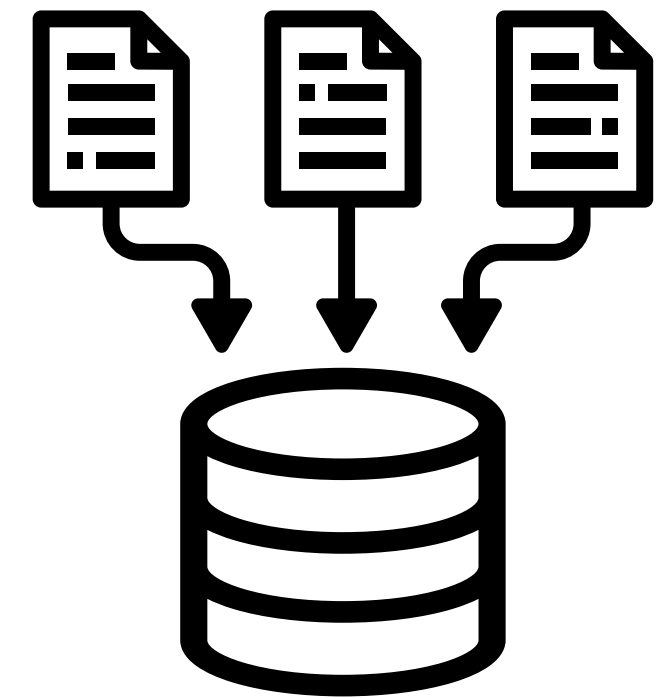
☐ Remember me

Sign In →

Don't have an account? [Create account](#)

LOGIN!

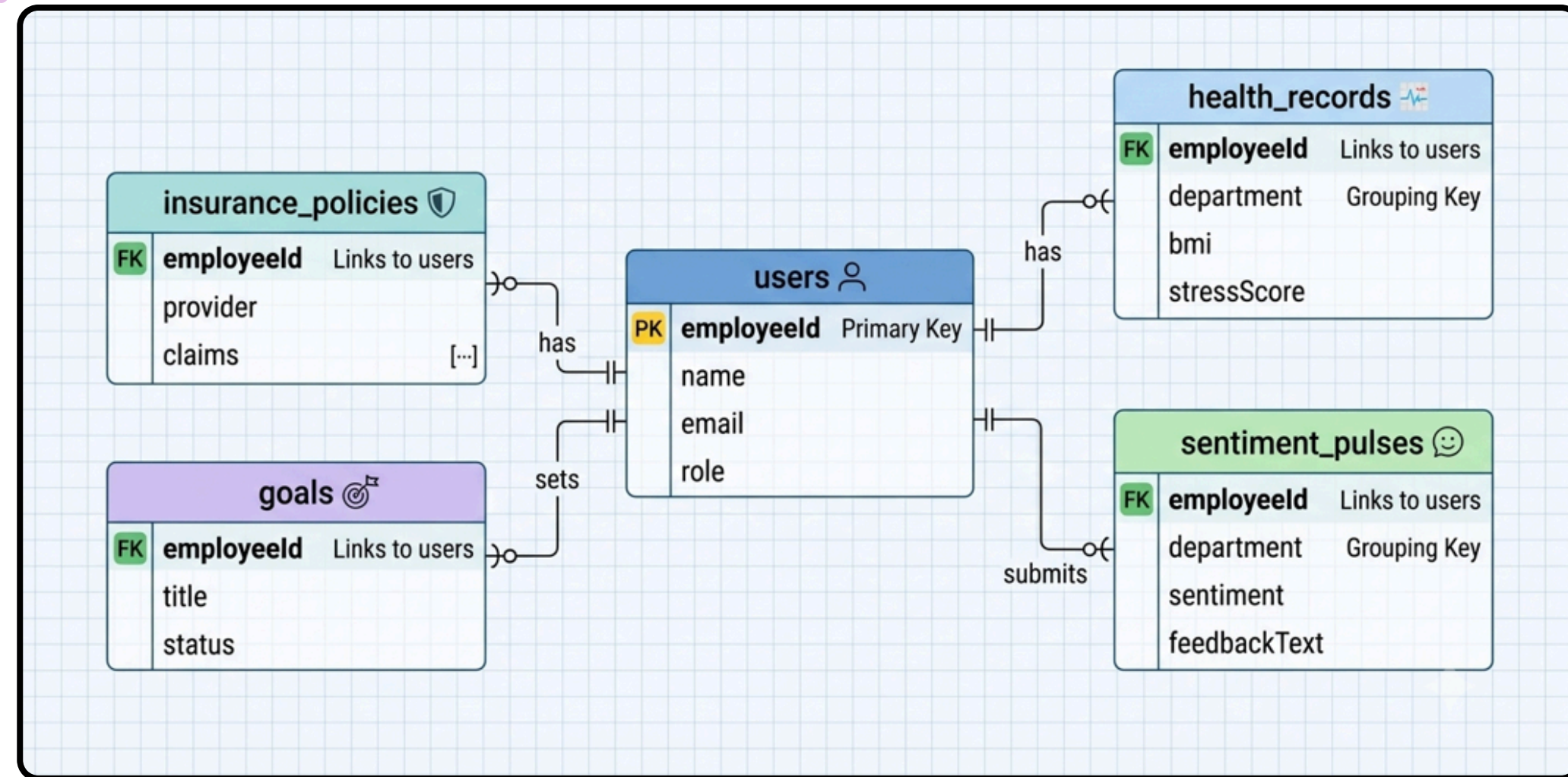
Database



➔ Database Management of the Employee Data

The backend connects to MongoDB using PyMongo.

User-related data is stored in collections including admin and employees in JSON – like format.



Health Records Collection



- Collecting the Health Record
- BMI and medical reports.
 - Exercise and sleep tracking.
 - Attendance and assessments.
 - Data preprocessing and validation.

Employee Health Data Management

Database logs for tracking key metrics including BMI, medical stats, sleep, and lifestyle routines.

Analytics Active

Search employee or ID...

All Departments

+ Add Employee's Health Profile

RI

Rishit Ranjan

EMP101

Dept: Engineering Age: 22

BMI: 23 Normal BP: 120/80

Ex (hrs/wk): 3 Sleep (hrs/nt): 8

Stress: Medium Glucose: 90

Condition: No major condition

Last Sync: Aug 2 2026 10:30 PM Fair

SU

Sudip Madhu

EMP102

Dept: Engineering Age: 19

BMI: 23.5 Normal BP: 120/80

Ex (hrs/wk): 3 Sleep (hrs/nt): 6

Stress: Medium Glucose: 78

Condition: No major condition

Last Sync: Jul 30 2026 5:30 AM Good

SH

Shiv

EMP103

Dept: Engineering Age: 21

BMI: 18.8 Normal BP: 120/80

Ex (hrs/wk): 0 Sleep (hrs/nt): 7.5

Stress: High Glucose: 90

Condition: No major condition

Last Sync: Jul 29 2026 5:30 AM Needs Attention

SI

Simran Hakim

EMP105

Dept: Engineering Age: 18

BMI: 24.2 Normal BP: 120/80

Ex (hrs/wk): 3.5 Sleep (hrs/nt): 3.5

Stress: Medium Glucose: 90

Condition: No major condition

Last Sync: Jul 28 2026 5:30 AM Fair

SA

Sanjana K Teggi

EMP104

Dept: Engineering Age: 20

Wellness Risk Prediction

→ Predicting the Risk

- Predicts stress and burnout.
- Detects obesity and hypertension risk.
- Uses historical and real-time data.
- ML: Decision Tree, Random Forest



MODULE 2 OF 5

Wellness Risk Prediction

Machine learning assessments predicting health risks, cardiovascular issues, or stress burnout.

HIGH SEVERITY

0 Score $\geq 70\%$

Critical risk indicators. Immediate clinical review or stress PTO mandated.

MODERATE SEVERITY

0 Score 45-69%

Elevated stress triggers. Guided meditation and ergonomic desk updates advised.

LOW SEVERITY

3 Score $< 45\%$

Healthy baseline. Maintain current lifestyle routines and claim fitness rewards.

Filter risk records by clinical severity:

All Risks (3)

High (0)

Moderate (0)

Low (3)

Rishit Ranjan

EMP101

LOW SEVERITY

Category

Risk Intensity Index: 42%

TRIGGERS DETECTED

Vitals check within ideal levels

PRESCRIBED ACTION

Low risk profile. Maintain current healthy routines and continue periodic monitoring.

Shiv

EMP103

LOW SEVERITY

Category

Risk Intensity Index: 41%

TRIGGERS DETECTED

Vitals check within ideal levels

PRESCRIBED ACTION

Low risk profile. Maintain current healthy routines and continue periodic monitoring.

Sudip Madhu

EMP102

LOW SEVERITY

Category

Risk Intensity Index: 30%

TRIGGERS DETECTED

Vitals check within ideal levels

PRESCRIBED ACTION

Low risk profile. Maintain current healthy routines and continue periodic monitoring.

Personalized Wellness Recommendation



Wellness Recommendation System

Tailored, evidence-based fitness routines, diet schedules, and mental wellbeing recommendations.

🔍 Search by employee name or ID...

Rishit Ranjan
EMP101

Low Risk



MENTAL WELLNESS

Guided Meditation Routine

10-15 minutes of guided mindfulness daily.

REASONS:

- Matches low-risk wellness profile



LIFESTYLE

Sleep Hygiene Program

Consistent bedtime routines and screen-free zones.

REASONS:

- Matches low-risk wellness profile



DIET

High-Protein Lean Macro Guide

Nutritional breakdown focusing on preservation of muscle tissue.

REASONS:

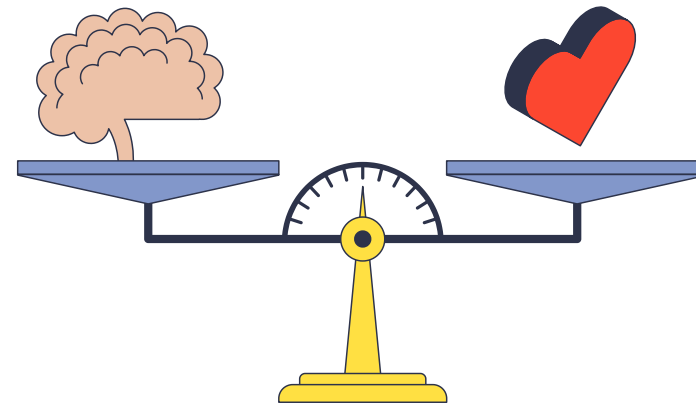
- Matches low-risk wellness profile
- Exercise can be improved further



Employee Wellness Recommendation

- Personalized fitness plans
- Diet and lifestyle suggestions
- Mental wellness and yoga activities
- AI: collaborative filtering, content-based filtering, Generative AI

Mental Health & Sentiment Analysis

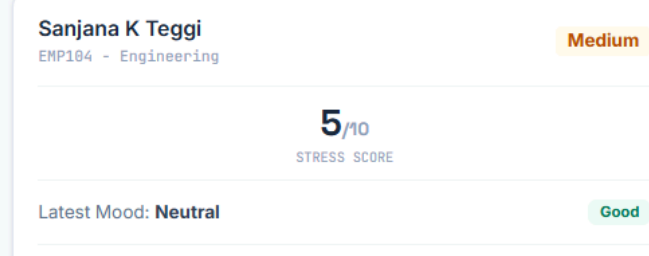
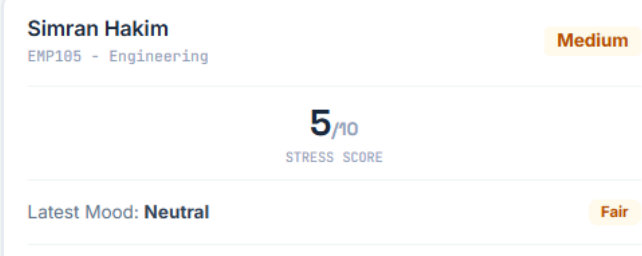
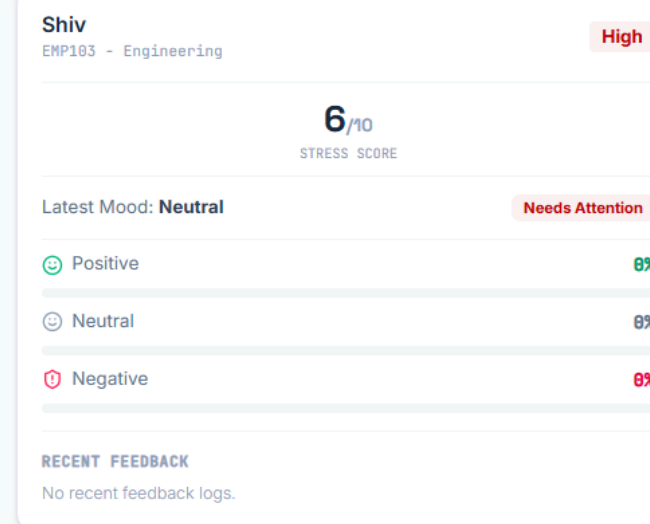
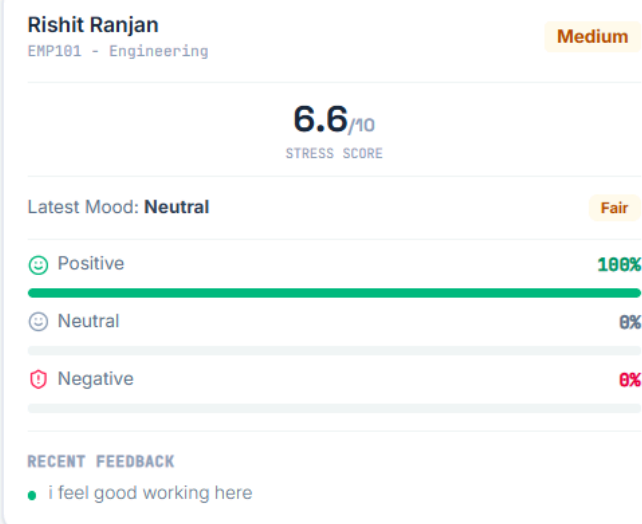


Mental Health & Sentiment Analytics

NLP-driven individual stress analytics collected through fully anonymized feedback pulse-checks.

Analytics Active

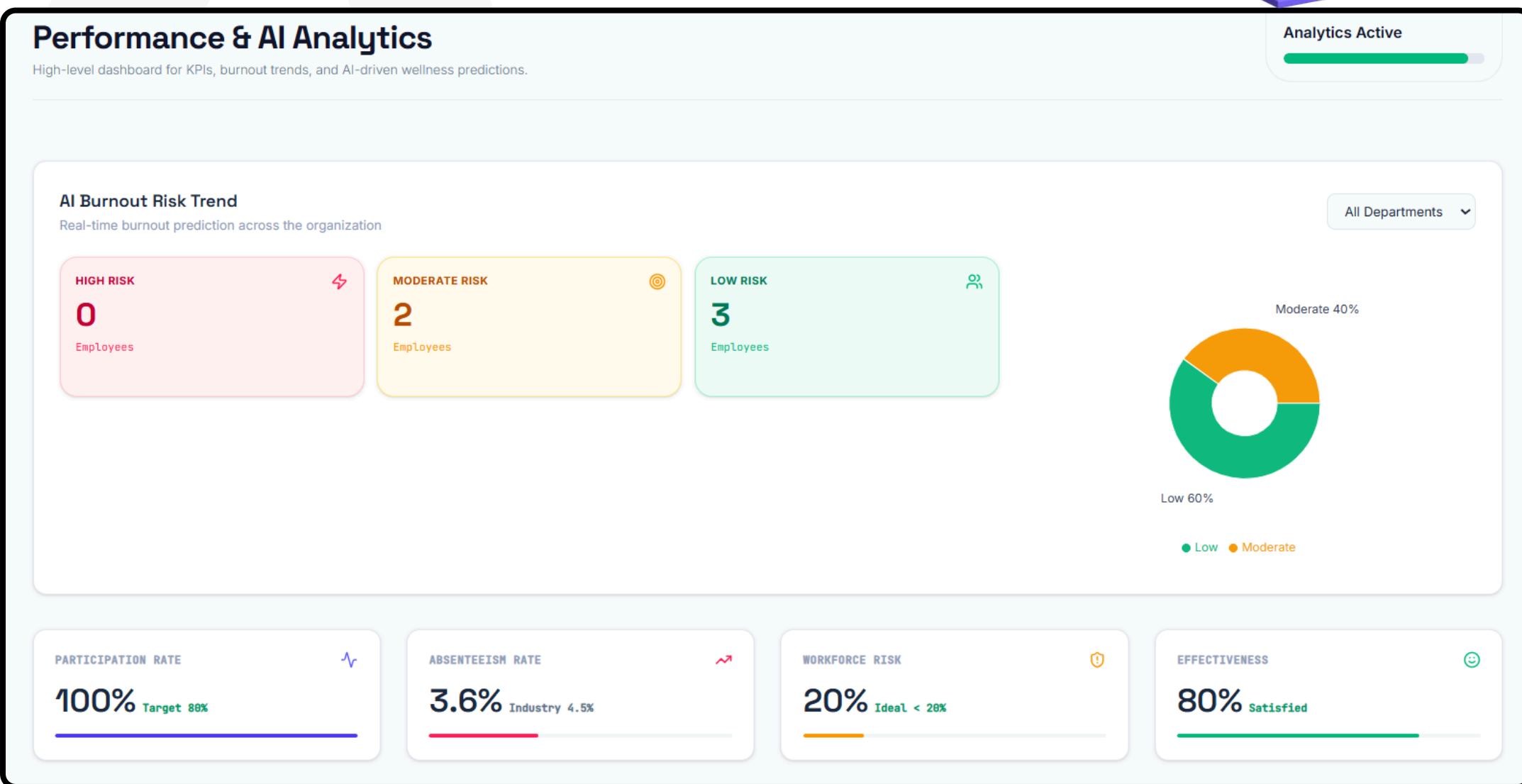
Individual Employee Sentiment



➔ Sentiment and Mental Health

- Analyzes surveys and feedback.
- Can use emails with consent and chatbot conversations
- Detects stress, anxiety, burnout.
- NLP, sentiment analysis, emotion detection

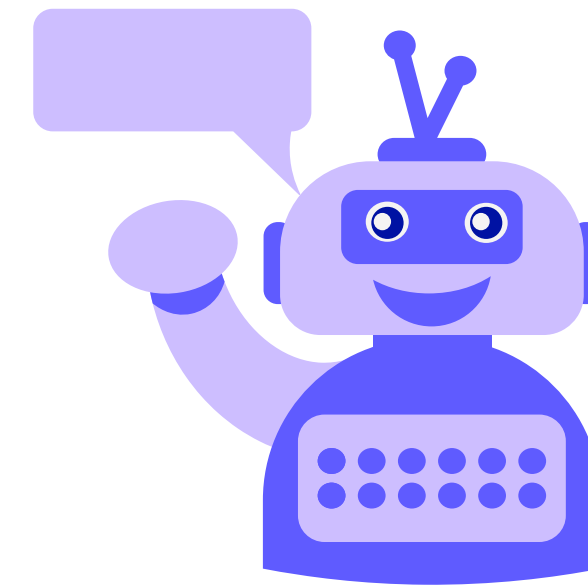
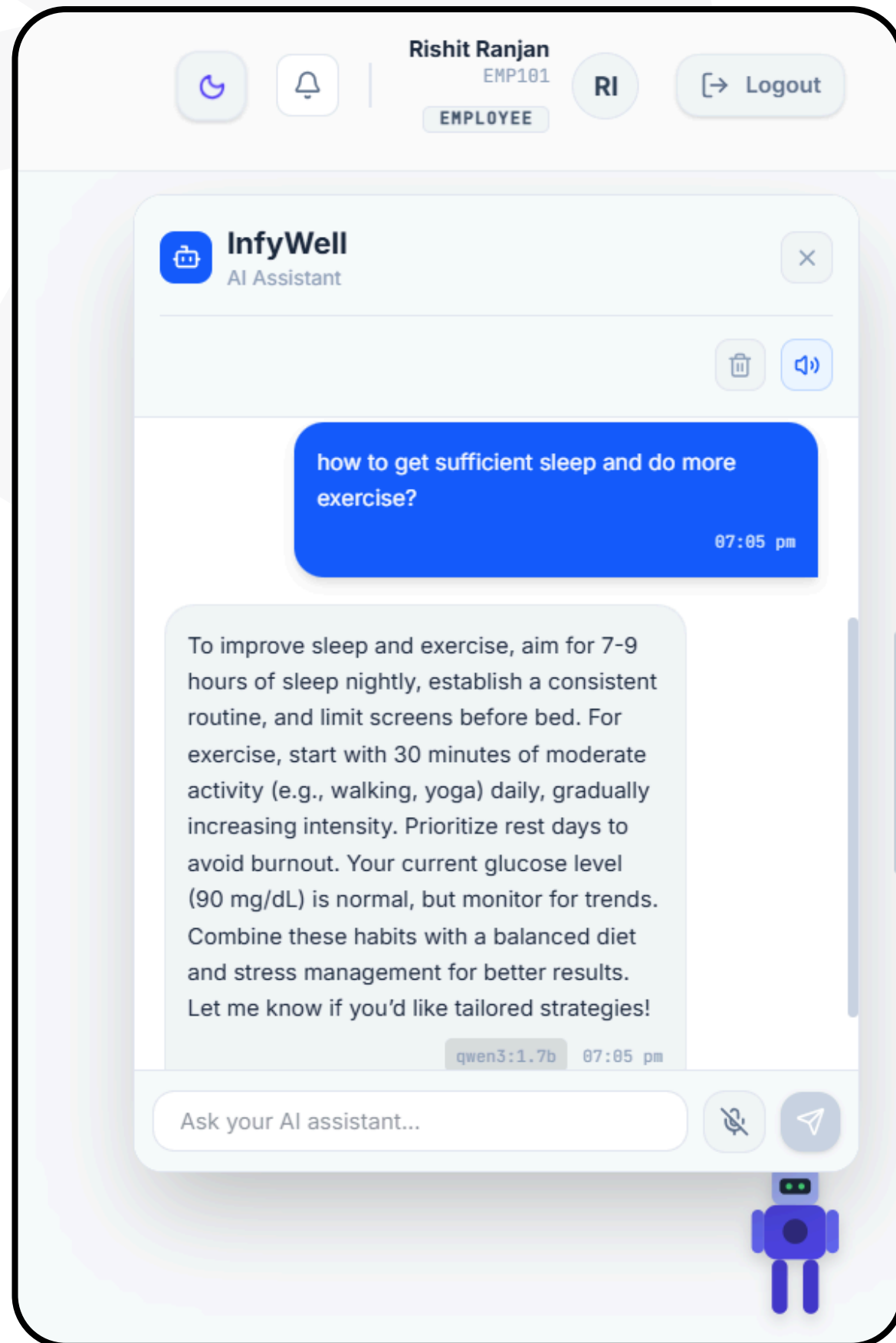
Performance & AI Analysis



➔ Employee Performance Metrics

- Tracks absenteeism and workforce health risk.
- Measures wellness program effectiveness.
- Helps admin monitor employee performance trends.

Chatbot Module



➔ Chatbot Support and Assistance

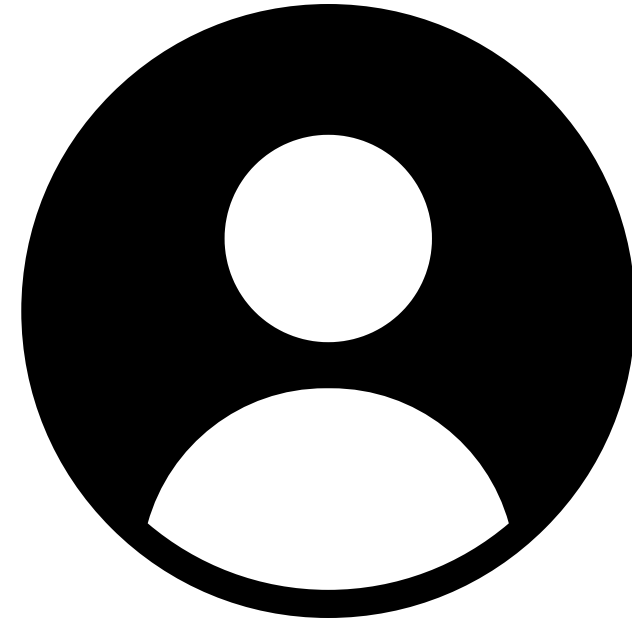
- AI-powered wellness assistant for employees.
- Answers questions on sleep, diet, exercise, stress, and burnout.
- Provides interactive chat support inside the user dashboard.



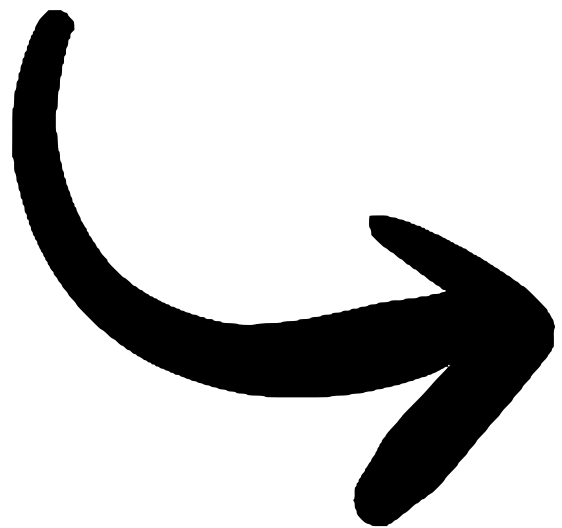
Additional

Features



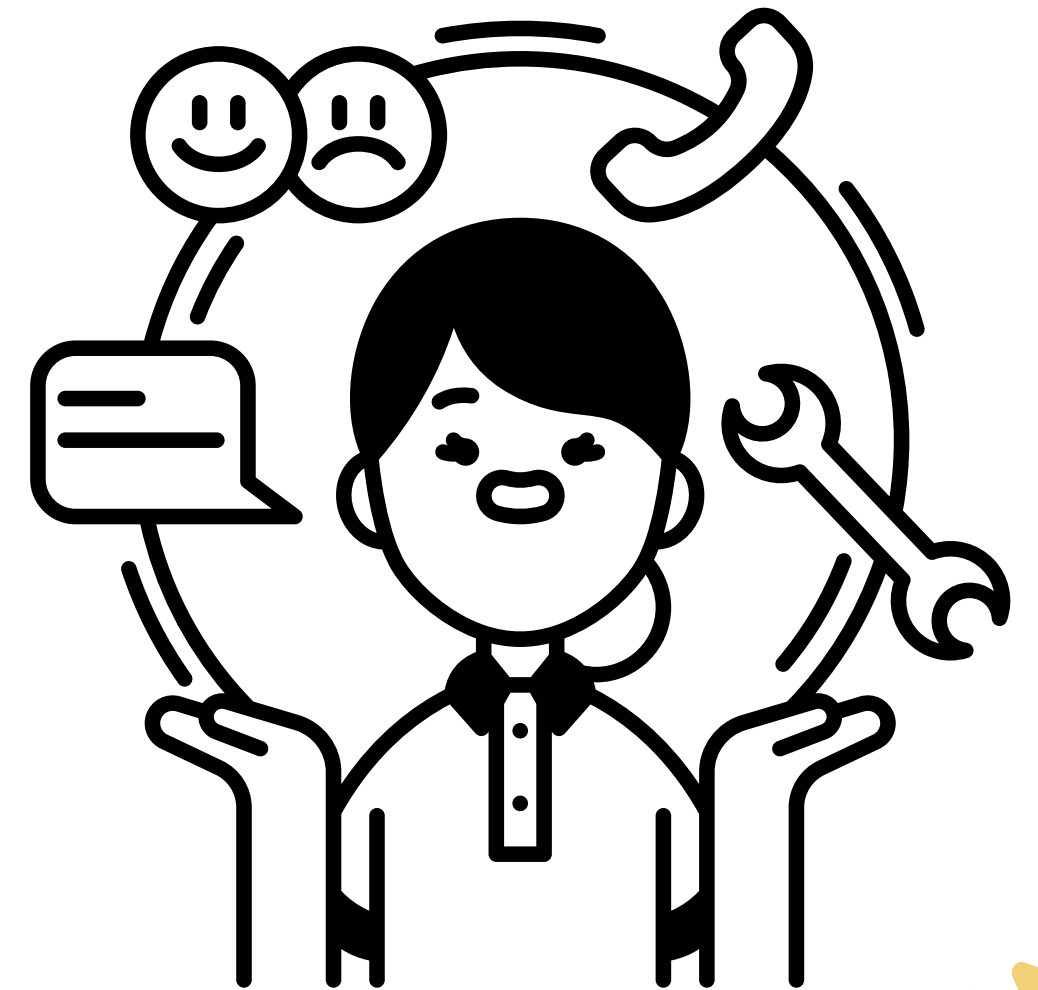


Admin Dashboard





Notification Center



Notification Center

Compose Notification

Title (e.g. Annual Health Camp)

General

Message...

Broadcast to all employees

Send

Sent Notifications

No notifications sent yet.

➔ Added Features for Analytics

- Broadcasting: Sends organization-wide announcements and updates to all employees simultaneously.
- Targeted Messaging: Delivers tailored notifications, wellness reminders, or private alerts to individual employees.



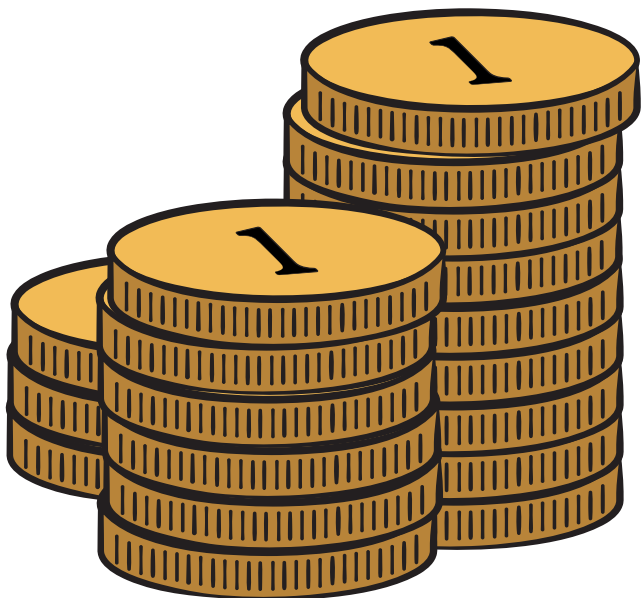
Checkups, SOS & Expenses

Checkups, SOS & Expenses				
Oversee employee checkup scheduling, SOS alerts, and expense claims.				
All Check-up Appointments				
Employee	Type	Date	Status	Actions
EMP103 (Shiv)	Annual Health Check-up	2026-07-23	Completed	Completed ▾
Emergency SOS Alerts				
Simran Hakim (EMP105) 27/7/2026, 8:31:38 pm Contact: — Phone: — Blood Group: — Allergies: —				Resolved
Sudip Madhu (EMP102) 27/7/2026, 7:21:08 pm Contact: — Phone: — Blood Group: — Allergies: —				Resolved
Shiv (EMP103) 24/7/2026, 8:42:57 pm Contact: — Phone: — Blood Group: — Allergies: —				Resolved
Shiv (EMP103) 22/7/2026, 8:43:50 pm				Resolved



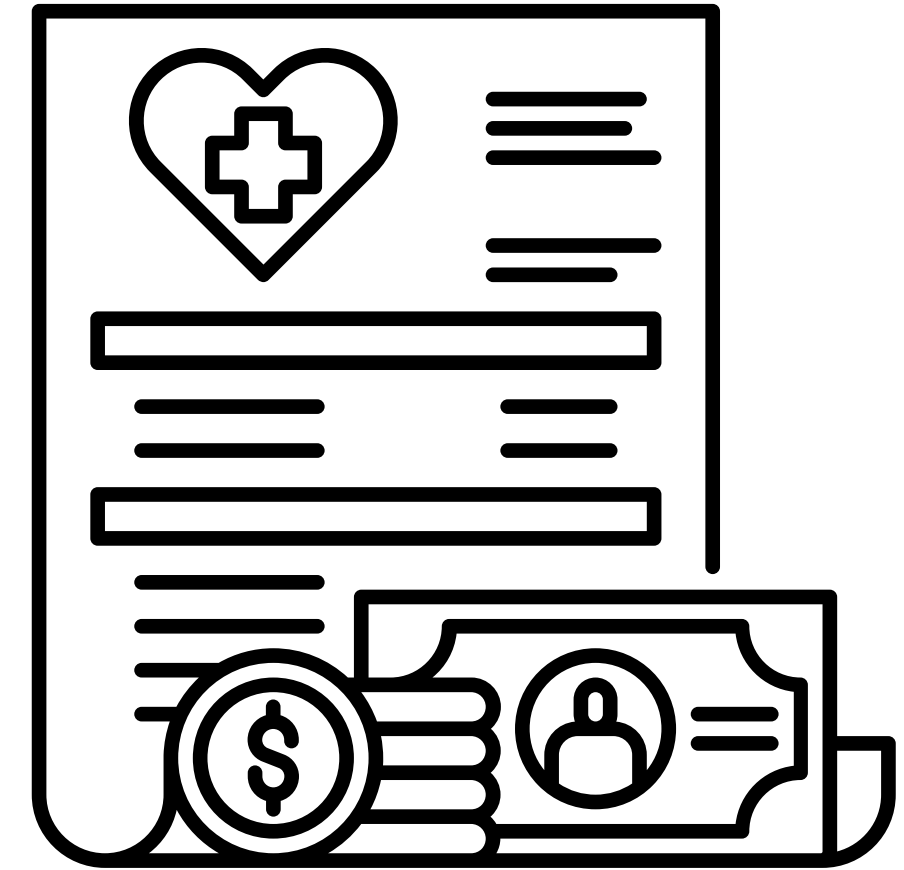
➔ Added Features for Analytics

- Emergency SOS Dispatch: Instantly alerts admins and emergency contacts with real-time location and vital logs during critical health events.
- Checkup & Expense Tracking: Simplifies routine checkup scheduling while centralizing health insurance claims and medical expense records.





Insurance Management



EXTENSION MODULE 6

Insurance Management

Manage employee insurance policies, claims, and coverage oversight.

+ Assign / Update Policy

EMPLOYEE	PROVIDER	POLICY NO.	COVERAGE (₹)	EXPIRY
Select...	Star Health			dd-mm-yyyy
Save Policy				

✓ Pending Claims (0)

No pending claims.

✓ All Policies

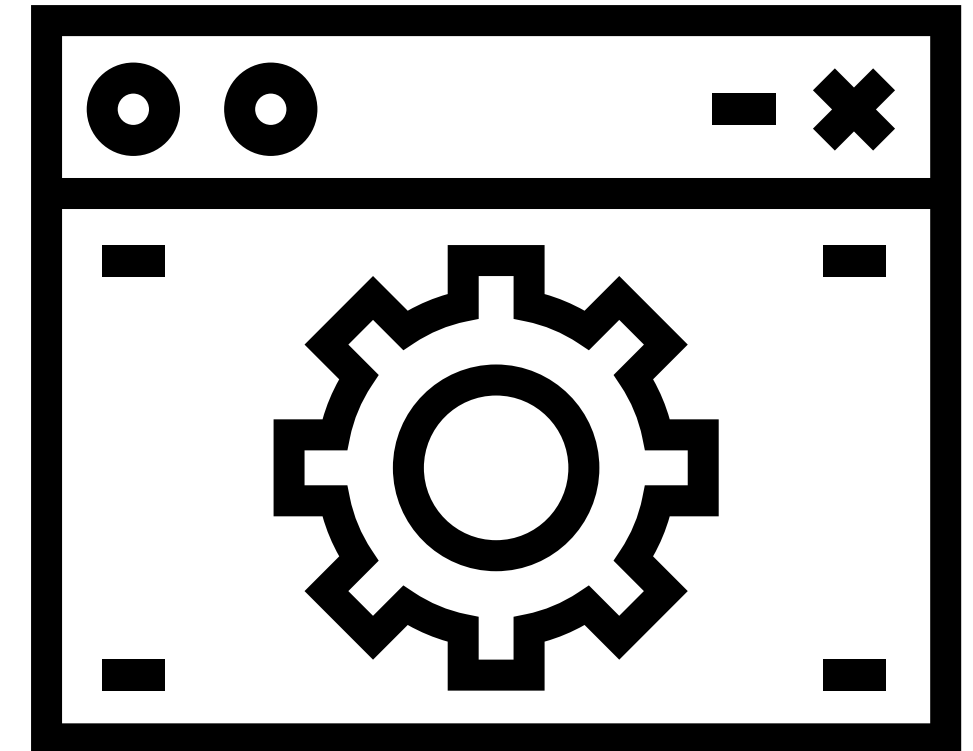
No policies assigned yet.

➔ Added Features for Analytics

- Claims Management: Monitors the status of submitted employee medical claims from filing to approval.
- Policy Monitoring: Tracks active insurance coverage, renewal dates, and individual benefit limits in real time.



System Settings



System Settings

Manage application-wide settings and configurations.

SYSTEM VITALS

Analytics Active



Risk Thresholds

Define the score boundaries for wellness risk categories.

High Risk Threshold (%)

70

Medium Risk Threshold (%)

45

Data & Privacy

Manage data retention and user privacy settings.

Data Retention (Days)

365

Anonymize Sentiment Data



Notifications

Manage how the system communicates with users.

Enable Email Notifications



Enable SMS Notifications

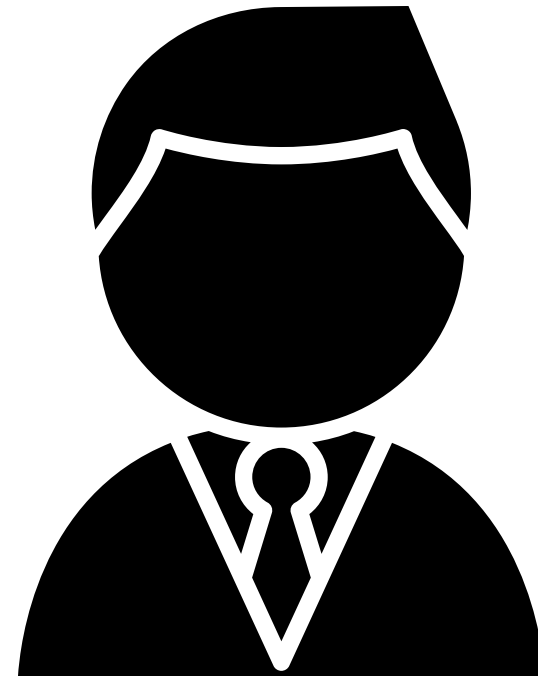


Save All Settings

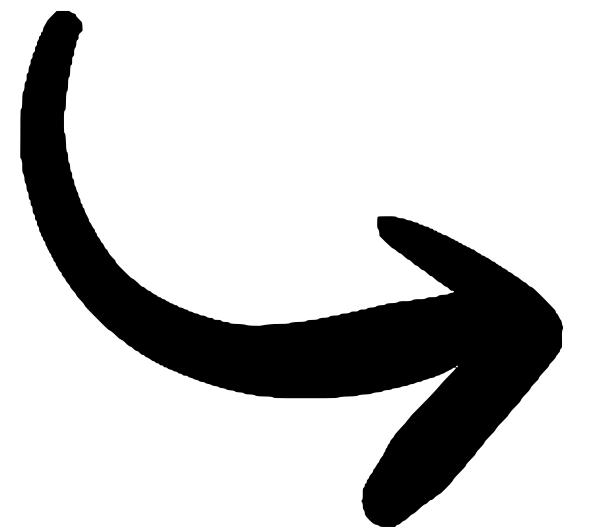


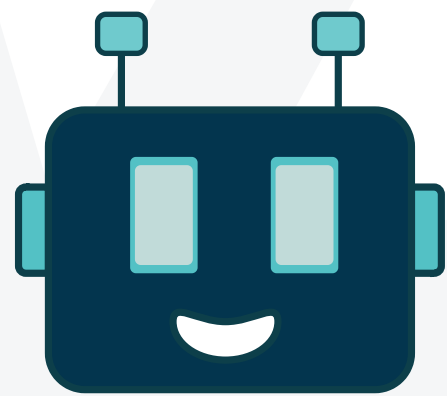
Added Features for Analytics

- **System Configuration:** Manages global application settings, feature toggles, and system-wide operational parameters.
- **Access & Security Rules:** Controls role permissions, notification preferences, and platform security policies.

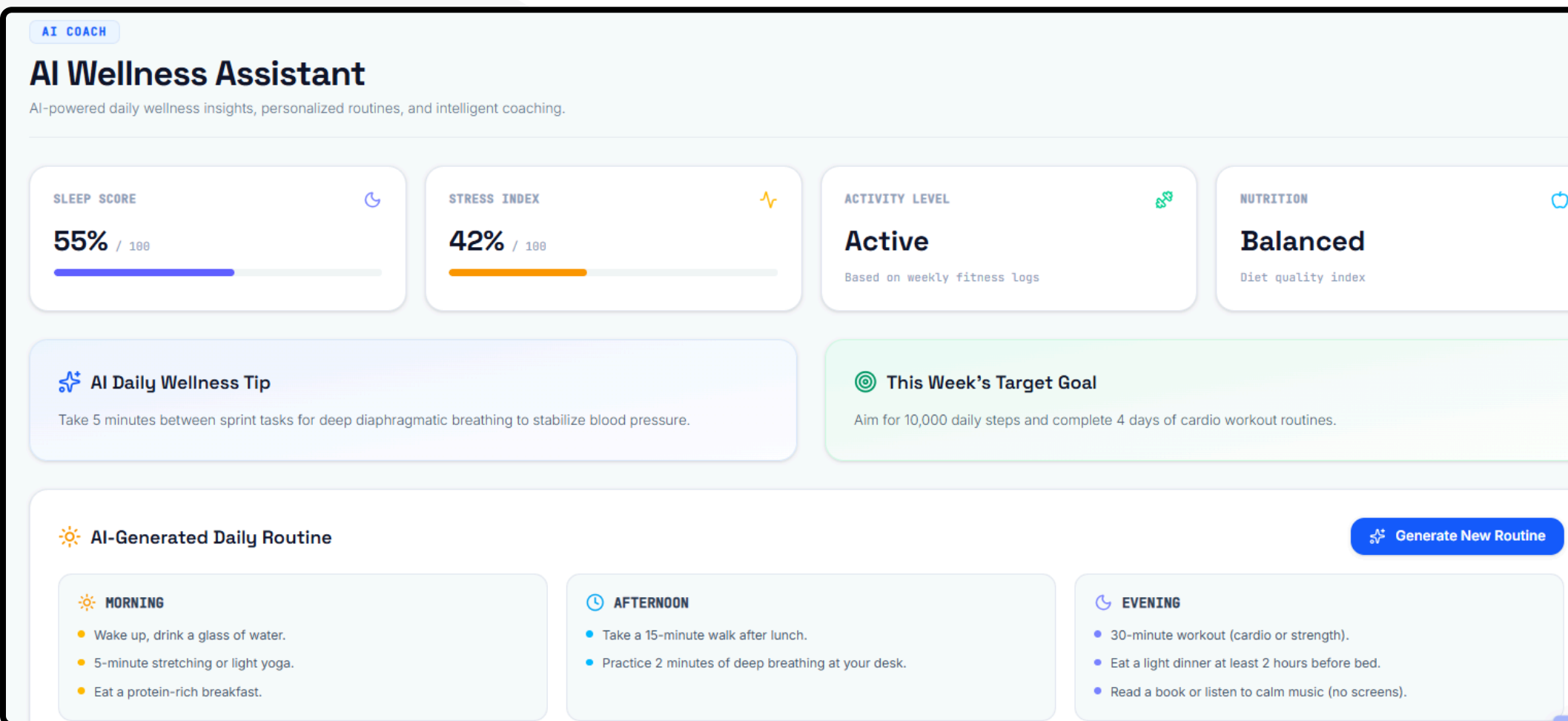


Employee Dashboard





AI Wellness Assistant



➔ Added Features for Wellness

- **Personalized AI Coaching:** Delivers real-time, tailored health recommendations and lifestyle guidance to employees.
- **Automated Support:** Provides 24/7 interactive assistance for daily wellness queries and mental well-being tracking.



AI Meals and Diet Plans



NUTRITION

AI Meal & Diet Plans

Personalized meal plans generated by AI based on your health profile.

CHOOSE YOUR DIET TYPE

Vegetarian Vegan Non-Veg Diabetic Weight Loss Weight Gain

CALORIES
1600-1800 kcal kcal

PROTEIN
65-75g

WATER INTAKE
3 L

Breakfast

- Vegetable Oats Upma with peas and carrots (1 bowl)
- Mixed Moong Sprouts Salad with lemon and cucumber (1 cup)

Lunch

- 2 Whole Wheat Chapatis
- Paneer and Mixed Vegetable Sabzi (1 bowl)
- Low-fat Cucumber Raita (1 cup)

Dinner

- Moong Dal and Spinach Khichdi (1 bowl)
- Stir-fried Broccoli and French Beans

Snacks

- Roasted Chana / Chickpeas (1 handful)
- 1 Apple with 5-6 soaked Almonds

This plan is a general guideline based on your selected diet type — adjust portions based on your activity level, and consult a nutritionist for medical conditions.



Added Features for Wellness

- Tailored Meal Plans: Generates customized diet plans matched to specific dietary types and user preferences.
- Preference Tracking: Dynamically adjusts nutrition goals based on individual health choices and dietary restrictions.





Health Reports and Timeline

REPORTS

Health Reports & Timeline

Download health reports as PDF and view your complete history.

Download Health Report

Generates a PDF snapshot of your current wellness score, BMI, stress, sleep, and recommendations.

Download PDF

Old Reports Timeline

Every time your health profile is updated, a snapshot is saved here so you can track changes over time.

No history yet — update your health profile to start building a timeline.



→ Added Features for Wellness

- PDF Report Generation: Exports comprehensive health summaries and wellness data into downloadable PDF formats.
- Instant Sharing: Allows employees to easily save, print, or share personal health reports with medical professionals.



Checkups & Emergency SOS

EMERGENCY

Health Checkup Scheduler

Schedule and manage your health checkup appointments.

Book a Check-up

DATE

dd-mm-yyyy

TYPE

Annual Health Check-up

NOTES (OPTIONAL)

+ Book Appointment

My Appointments
No check-ups booked yet.



FINANCIAL

Emergency SOS History

Review the history of all emergency SOS alerts you have triggered.

Emergency SOS
Pressing this immediately alerts the admin/HR team with your emergency contact and known health info (blood group, allergies, conditions).

Trigger SOS Alert

My SOS History
No alerts triggered.

➔ Added Features for Wellness

- **Automated Checkup Scheduling:** Streamlines health checkup bookings directly through the user and admin dashboards.
- **Emergency SOS Trigger:** Instantly dispatches high-priority alerts with real-time location and vitals to emergency contacts during medical crises.



Conclusion

- The project includes secure authentication, database integration, risk prediction, recommendation, sentimental analysis, performance analytics, and an AI wellness chatbot in one platform.
- It supports both employee and admin roles for managing wellness data and system features.
- These modules make the system more intelligent, interactive, and useful for future expansion.
- Modular Scalability: Designed with an extensible architecture that allows seamless integration of future health-tech tools and wearable devices.
- Real-time Incident Response: Combines continuous health metric tracking with automated SOS triggers and expense management for rapid emergency handling.

**Thank you
so much**

